

Exclusive Guest-Physical Isolation in a Type-1 Hypervisor: A Machine-Checked Ghost Model and a Bare-Metal Runtime

Anonymous Author(s)

Submission Id: eurosys27-fall-anon

Abstract

Type-1 hypervisors isolate guest memory with extended page tables, yet production stacks treat that isolation as a testing problem. This paper reports Isolon, a clean-slate Type-1 hypervisor that boots as a single UEFI binary on production two-socket Xeon servers and is designed so the security-critical path can be machine-checked.

Isolon confines formal proofs to a Proven Core. The headline property is *exclusive ownership* of host frames: every valid guest-physical to host-physical mapping is owned by exactly one guest and belongs to neither the hypervisor nor any other guest. We discharge that property in a host-only Verus ghost model through map, unmap, EPT-violation handling, and a page-transfer primitive—80 lemmas verified, 0 errors, with no admit on the exclusivity path—under a frozen SMT toolchain pin. Live executable EPT remains at spec-written maturity: a runtime ownership registry plus assertions, not a bisimulation of the EFI. Device emulation, the management plane, and guest firmware are outside the Proven Core and are not claimed as theorems.

On a real PowerEdge R640 we reproduce VMX bring-up, EPT identity mapping, an unmodified Linux shell, multi-VM probes, durable host-NIC HTTP, and guest-firmware CD-ROM boot. Nested VT-x completed a Linux distro install-to-disk; nested virtualization is not that server. We report those facts, name the ghost/executable gap, and stop there.

CCS Concepts: • **Software and its engineering** → **Software verification**; **Virtual machines**; • **Security and privacy** → *Logic and verification*.

Keywords: formal verification, Type-1 hypervisor, EPT, Verus, exclusive ownership

ACM Reference Format:

Anonymous Author(s). 2027. Exclusive Guest-Physical Isolation in a Type-1 Hypervisor: A Machine-Checked Ghost Model and a Bare-Metal Runtime. In *Proceedings of European Conference on Computer Systems (EuroSys '27)*. ACM, New York, NY, USA, 8 pages.

1 Introduction

A hypervisor’s job is to make one computer look like several, without letting those several computers look at each other. On modern Intel servers the mechanism that actually enforces that is Extended Page Tables (EPT) [8]. If EPT is wrong, two guests share a frame, or a guest sees hypervisor

memory, and no amount of management-API polish repairs it.

Commercial Type-1 stacks—ESXi, Xen [1], KVM [9]—are large, mature, and tested. They are not accompanied by a machine-checked isolation theorem for the frames they hand to guests. Research systems that *are* proved (seL4 [10, 11], CertiKOS [5], Komodo [4]) target microkernels, certified kernels, or enclaves. They are not a single UEFI binary aimed at the PowerEdge fleet already in the rack.

Isolon takes a narrower bet. It is a clean-slate Type-1 hypervisor written in Rust. It compiles to one `.efi`. It boots on hardware we own and can serial-log; Dell PowerEdge R640-class machines. Formal verification is an architectural constraint, not a post-hoc whitepaper. We picked Verus [12, 13] for SMT proofs of ghost state and Kani [21] for bounded checking of unsafe, and we defined a four-level maturity ladder so the project can ship at runtime-enforced or spec-written maturity when a full proof is still in flight.

The headline theorem is exclusive ownership of host frames. We prove it of a *ghost model*, not of the EFI that leaves the compiler. That distinction is the paper. Claiming “the hypervisor is proved” would be proving the wrong thing [10]: a convenient proxy. We discharge 80 Verus lemmas with 0 errors; live executable EPT is a 705-line ownership registry with contracts and assertions. Device emulation, HTTP, the SPA, ISO handling, and guest firmware are outside the Proven Core.

This paper makes three contributions:

1. **A tight isolation theorem and a discharged ghost model.** Exclusive ownership is four conjuncts over two ghost maps (frame $\rightarrow$ guest, and (guest, GPA) $\rightarrow$ frame). Named theorems cover 4K map/unmap (one guest and N guests), scoped ghost $\leftrightarrow$ executable refinement, 2M/1G spans, NUMA affinity, hardware 2M-leaf decode, EPT-violation dispositions, and a page-transfer primitive. Under a frozen Verus pin the crate reports 80 verified, 0 errors, with no admit on that path.
2. **An architecture that keeps the proved surface small** while still shipping a UEFI Type-1 hypervisor: a 15,000-line Proven Core budget, a host-only proof crate that is deliberately not linked into the `.efi`, and an explicit refusal to mutate third-party firmware internals to manufacture forward progress.

3. **A split evaluation.** Lab/CI hosts hold L3. A real PowerEdge R640 holds L1 runtime through Linux shell, multi-VM probes, host-NIC HTTP, and guest-firmware El Torito. Nested VT-x completed a distro sysinstall. We do not collapse those three columns into one claim.

Isolon is not seL4 for hypervisors and is not a verified KVM. It is a small exclusive-ownership model that we actually discharge, sitting next to a hypervisor we actually boot on iron, with the gap named in the same breath.

2 Threat Model and Non-Goals

2.1 Why memory isolation is the headline

A Type-1 hypervisor is the TCB for every guest it runs. CPUID and MSR firewalls, interrupt injection, and the hypercall door all matter. None of them matter if EPT can map the same host frame to two guests, or to a guest and the hypervisor, without anyone noticing. We therefore name *exclusive ownership of host frames* as the property we prove, not “the guest seemed isolated in tests.”

2.2 In scope

We assume:

- **A malicious or compromised guest** (ring-0 Linux, firmware, or installer) that may issue any architected VM exit: EPT violation, CPUID, MSR, I/O, hypercall, HLT, exception.
- **Buggy device emulation** outside the Proven Core (virtio, IDE/ATAPI, UART). Bugs there can DoS a guest or confuse an installer; they must not become a second EPT mapping of a Proven Core frame.
- **Operator error** (wrong ISO, exhausted USB stick). Audit events record what happened; they do not enlarge the theorem.

We do *not* assume a correct third-party guest firmware. Forcing a foreign firmware’s WaitForEvent internals to skip a check produced a NULL-event page fault and a dead loop. Product ISO boot now uses firmware we author. That firmware is **outside** the Proven Core.

2.3 Out of scope for the theorems

- The Intel SDM’s VMX/EPT hardware behaves as documented [8].
- The frozen Verus/SMT stack is trusted for the ghost proofs.
- Host physical memory given to the allocator is the memory the machine has.
- DMA from devices we do not model (untrusted NICs, USB) is a future confinement problem, not a discharged lemma.
- Nested VT-x on a lab server is not a proof about a PowerEdge R640.

Table 1. Proven Core budget (cap, not an L3 claim per row).

Module	Est. LOC	Why inside
VMX lifecycle	800	VMXON/VMXOFF leave CPU mode
VMCS management	1,500	Host-state corruption
EPT engine	2,000	Isolation mechanism
Frame allocator	1,500	Double-alloc / UAF
vCPU state	1,000	Save/restore leaks
Interrupt injection	800	Guest privilege
Hypercall door	500	Only intentional channel
MSR/CPUID/CR firewalls	1,200	Host feature subversion
Audit-log integrity	600	Tamper-evident trail
IPI confinement	500	Cross-VM IPI
Cap (code + scaffolding)	15,000	—

- Full x86 ISA / VMCS host-state correctness is inside the Proven Core *budget* and is not at proof-complete maturity beside the EPT crate.

3 Isolon Architecture

3.1 Single binary

Everything compiles to one UEFI PE/COFF image. Kernel, initrd, Web UI, and schemas are compressed sections, decompressed on first use. The size target is 15 MB with a 20 MB hard limit. There is no host Linux, no package manager, and no config tree the operator must get right before the first serial line.

3.2 Proven Core boundary

Only the modules in Table 1 are in scope for Verus specifications and progressive proofs. Everything else uses Rust’s type system, testing, and review. Promotion into the Proven Core requires a written architecture decision; the default answer is no. The hard cap is 15,000 lines including proof scaffolding.

Lived split, as of this submission: the machine-checked exclusivity theorems live in a host-only crate (ept_model, 2,366 lines) that is *not* linked into the .efi. The executable EPT ownership registry is 705 lines at spec-written maturity. Device emulation, HTTP, the SPA, ISO, guest firmware, iDRAC, and migration tooling are outside.

3.3 Hardware

Dell PowerEdge R640 / R650 / R660. Low-risk, self-sufficient surfaces (serial via iDRAC virtual console, SMBIOS/ACPI topology, onboard LOM) are in scope. OEM RAID health schemas are not a blocker for the theorems or for the single-host product loop.

Code in the Proven Core is required to document invariants at the function boundary, keep unsafe blocks minimal, and tag each of them for bounded checking. Mutable globals require a lock and a justification comment. We use explicit

state machines rather than boolean flags. These rules are how a codebase stays L3-shaped when most modules are still L1: the later proof should not require a rewrite of ownership.

3.4 Maturity ladder

We use four levels, kept even after a proof lands (runtime asserts stay):

- **L0** documented invariants in comments.
- **L1** runtime `assert! / debug_assert!` plus Kani on `unsafe`.
- **L2** Verus specifications with ghost state and contracts.
- **L3** cargo `verus verify` succeeds for the claimed surface.

Fallback is `Verus` $\rightarrow$ `Kani` $\rightarrow$ `asserts` and `fuzz`. The architecture always aims at L3; shipping is allowed at L1/L2 when L3 is still open, provided the paper does not upgrade the label.

Proven Core modules follow a four-file layout: executable Rust, a spec file, a proof file, and tests (Kani + unit). The exclusivity theorems of record do not use that layout. They live in one `verus!` block in `ept_model`, verified with `cargo verus verify`. Putting them beside live memory/`ept.rs` would imply the EFI crate is L3. It is not.

4 Exclusive Ownership

4.1 Statement

For every valid EPT mapping from a guest-physical address to a host-physical frame, that frame is exclusively owned by the mapping guest and belongs to neither the hypervisor nor any other guest.

“Exclusively owned” means exactly one guest holds a mapping to that frame at any moment. “Belongs to neither” means the frame is absent from the hypervisor’s page tables and from every other guest’s EPT. The property must hold across `map`, `unmap`, EPT-violation handling, and live-migration page transfer.

4.2 Ghost state

The ghost model is two finite maps (Figure 1):

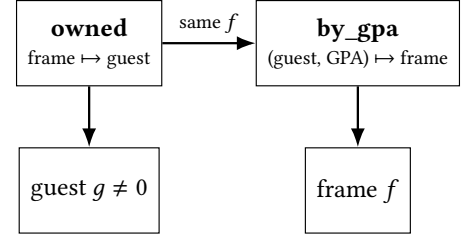
- $owned : FrameId \rightarrow GuestId$
- $by_gpa : (GuestId \times Gpa) \rightarrow FrameId$

Guest id 0 is reserved (hypervisor / unmapped). 4K steps require page-aligned GPAs. A `map` step is enabled only when the frame is free and the (guest, GPA) pair is free. An `unmap` step is enabled only when that pair is present.

The predicate (from the crate, names only) is:

`exclusive_ownership(m)` iff every owned frame has a `by_gpa` witness, every `by_gpa` entry has a matching owner, a frame is the image of at most one pair, and the two maps have equal length.

Empty maps satisfy it. A hypervisor frame is modelled as *not* being in `owned`: the ghost registry only tracks guest-held



Bijection: equal cardinalities; no two pairs share a frame unless they are the same pair.

Figure 1. Ghost EPT ownership registry. Exclusive ownership is four conjuncts: every owned frame has a matching `by_gpa` witness; every `by_gpa` entry has a matching owner; a frame is the image of at most one pair; the two maps have equal length.

frames. That is the “belongs to neither” half, as a modelling choice—the EFI’s own page tables are not in this crate.

4.3 Steps

A 4K step is $\text{Map}\{g, a, f\}$ or $\text{Unmap}\{g, a\}$. Enabled steps preserve exclusive ownership (one-step lemma). Any finite sequence of enabled steps preserves it (the historical single-guest theorem, whose statement already allows distinct guest ids across steps; a second theorem is named so CI can assert the N -guest post). A two-guest lemma maps two distinct non-zero guests onto distinct free frames.

Large pages are spans of 4K frames (512 for 2M, 2^{18} for 1G). Span `map/unmap` lemmas lift exclusivity to 2M/1G; unit lemmas record that 4K is a one-frame span. NUMA lemmas couple a ghost SRAT/SLIT topology with `map/unmap` so a mapped frame is recorded on its node. Hardware 2M-leaf decode lemmas relate a leaf PTE encoding to an identity-map intent. None of these lemmas are a full multi-level EPT walk of the EFI.

4.4 EPT-violation dispositions

An EPT miss is modelled as one of three dispositions:

- `EmulateNoMap` — emulate MMIO (APIC / virtio) without installing a mapping. Registry unchanged.
- `Reject` — fail closed. Registry unchanged.
- `ClaimMap` $\{g, a, f\}$ — demand-fill a 4K map, enabled only when that map step would be enabled.

Every enabled disposition preserves exclusive ownership. `Emulate-then-claim` is a derived post.

4.5 Page transfer

Live migration of a frame is `unmap-at-source` then `map-the-same-HPA` at destination. The step is enabled when source holds the GPA, destination GPA is free, and guests are distinct and non-zero. Posts: destination owns the frame at the new GPA; the source mapping is gone; exclusivity holds. This is a

ghost primitive. It is not vCenter vMotion, not a multi-host protocol, and not a proof that the product migrate engine is correct. That engine is outside the Proven Core.

4.6 Theorems of record

Table 2 lists the named theorems. Supporting lemmas (empty-exclusive, map-already-owned-unchanged, span preservation, mock bring-up facts) are in the same crate. CI counts 80 verified functions.

4.7 Proof sketch: map preserves exclusivity

We sketch the 4K map case; unmap is the inverse bookkeeping. Assume *exclusive_ownership*(m), $g \neq 0$, a 4K-aligned, $f \notin \text{dom}(\text{owned})$, and $(g, a) \notin \text{dom}(\text{by_gpa})$. Let $m' = m.\text{ghost_map}(g, a, f)$.

Witness. For the new frame f , the pair (g, a) is the witness required by conjunct 1. Old frames keep their old witnesses because we did not delete *by_gpa* entries.

Owner match. The new *by_gpa* entry points at f , and $\text{owned}'[f] = g$. Old pairs are unchanged.

Uniqueness. Suppose two pairs in m' share a frame. If that frame is f , both pairs must be (g, a) because f was free. If not, uniqueness in m applies.

Cardinalities. Both maps gain one key, so lengths stay equal.

Verus discharges this with map lemmas from *vstd* (broadcast *group_map_lemmas* / *group_set_lemmas*). The one-step lemma case-splits on Map vs Unmap. The sequence theorem is induction on Seq length: the empty sequence is trivial; the inductive step applies the one-step lemma and recurses. The N -guest theorem is that same induction under a CI-facing name; guest ids are already a field of each step.

What the SMT does not get for free. Span (2M/1G) map is not “map 4K k times” as an opaque loop in executable code. It is a ghost predicate that the 4K keys covering the span are all free, then a bulk insert. The 2M lemma records that 512 aligned 4K frames constitute a 2M leaf; the 1G lemma records 2^{18} . Those facts are arithmetic lemmas, not a hardware page-walk.

Violation ClaimMap reuses the 4K map lemma. Emulate and reject are identity on the registry; the proof is $m' = m$. Transfer is unmap then map of the *same* frame: after unmap, f is free, so the destination map is enabled. Posts are the destination owner and the absent source key.

Modelling choice: hypervisor frames. Conjunctions quantify over *owned* and *by_gpa*, not over the EFI’s own page tables. A hypervisor frame is exclusive in this model because it is *absent* from *owned*. That is the right theorem for “guests do not share guest frames.” It is *not* a theorem that an executable bug cannot install a guest leaf pointing at `.text`. Closing that would require the live EPT walker and the HV page tables in the same proved artifact. They are not. Section 5 is that sentence at L2.

Table 2. Named exclusivity theorems in the host-only crate.

Theorem	Claim
<code>single_guest_4k_map_unmap</code>	finite 4K sequences
<code>n_guest_4k_map_unmap</code>	same discharge, N -guest name
<code>concrete_*_guest_refine</code>	scoped $\text{ghost} \leftrightarrow \text{exec}$
<code>large_page_map_unmap</code>	2M/1G spans
<code>numa_map_unmap_affinity</code>	node affinity ghost
<code>hw_2m_leaf_refines_</code>	2M leaf encode
<code>ept_violation_preserves</code>	miss dispositions
<code>page_transfer_preserves</code>	unmap $\rightarrow$ map handoff
<code>alloc_map_unmap_refines</code>	pool + EPT refine

5 The Ghost / Executable Gap

A discharged ghost model is not a proved .efi. Isolon records that as a first-class result, not a footnote.

Live EPT (`memory/ept.rs`, 705 lines) maintains an ownership registry and asserts exclusive-ownership-shaped facts at map/unmap. Beside it sit 357 lines of spec/proof notes. Those notes are not the 80-lemma crate. The host-only crate exists *because* linking Verus into the UEFI target is the wrong packaging: proofs run on a Linux CI host under a pinned Verus release; the EFI is a UEFI target, not a Linux sysroot.

Scoped refinement lemmas show that a concrete map representation, under a restricted interpretation, commutes with ghost map/unmap for 4K (one guest, then N guests) and that an allocate-then-map path refines. They are not a bisimulation of VMXON, VMCS host state, EPT pointer load, and every MMIO emulator the EFI contains.

Consequences we accept in print:

- L3 is the ghost crate. Live EPT is L2.
- VMX, VMCS, hypercalls, MSR firewalls, audit integrity, and IPI confinement are in the budget in Table 1 and are not at L3 beside `ept_model`.
- Iron serial logs raise confidence in the *runtime path*. They do not raise Verus maturity. Lab L3 and R640 L1 are both real; they are not the same claim.

A prior self-review asked whether exclusive ownership is the property an auditor cares about or a convenient proxy. The verdict was: in scope for map/unmap/violation/transfer; ops gates (auth, PDF reports, HA, soak metrics) correctly labelled outside. Independent third-party re-run of `verus-verify` under the pin remains the auditor path. Bring-up self-review is not a conference peer review.

5.1 Live registry versus ghost maps

The live engine is not a Verus Map. It is a bounded table: `EptMap` holds at most 512 explicit 4K claims in production builds (8 under Kani, so CBMC can unwind), plus a 16-slot range registry for the precise identity window used at Linux bring-up. Map fails with `AlreadyOwned` when a frame is

taken. Invariants are checked with `debug_assert!` after mutating operations. A boot self-test maps a bring-up guest, then asserts that a second guest cannot alias the same frame.

That is the right shape for L1/L2. It is a different datatype from the ghost model (finite maps of unbounded domain, no `MAP_CAP`). The scoped refine lemmas interpret a concrete map into ghost maps; they do not prove that the 512-slot table implements those maps for every reachable EFI state. If a 513th distinct 4K claim is required, the live table saturates while the ghost theorem still holds of an abstract `ghost_map`. That mismatch is why we will not stamp L3 on `memory/ept.rs` from the 80/0 line.

The range registry is another gap. Bring-up identity-maps a contiguous window (hundreds of megabytes) without inserting hundreds of thousands of 4K ghost keys at runtime. The ghost large-page lemmas talk about spans; the live precise window is a range claim plus hardware identity PTEs. Correspondence is scoped (hardware 2M-leaf decode lemma + range self-test), not “every live PTE has a ghost key.”

5.2 What would close the gap

Three artifacts, none of which this paper claims:

1. A proved executable EPT walker whose leaves are in 1–1 correspondence with `by_gpa`, including the HV-vs-guest split (“belongs to neither” as a fact about HV page tables, not as absence from a guest registry).
2. An IOMMU / DMA story: devices we do not model cannot write a guest-owned frame or a hypervisor frame.
3. Linking or replaying (1) on the UEFI target, or a refinement from the EFI binary’s EPT root pointer to the ghost maps.

Until then, “memory isolation isn’t tested; the ghost exclusivity theorems are proved” is the strongest true slogan. “The hypervisor is proved” is false.

6 Evaluation

We evaluate three questions. (1) Does the ghost model actually verify? (2) How large is the proved surface relative to the hypervisor? (3) What does the runtime do on a lab host versus a real R640?

6.1 Proofs

Toolchain: Verus pin 0.2026.07.12 (asset 0b42f4c; content-addressed Linux zip, SHA-256 pinned; CI never uses latest) and Kani 0.67.0. The exclusivity crate is verified with cargo verus verify. The last closed host gate on that crate reported **80 verified, 0 errors**. No admit on the exclusivity path. We do not invent a wall-clock for this PDF; the artifact is the crate plus the pin (supplement).

Kani covers bounded unsafe in the live engine. That is L1 on unsafe, not L3 on exclusive ownership.

Table 3. Line counts (wc -l) for the EPT-related surface.

Path	Lines	Role
<code>ept_model crate</code>	2,366	L3 ghost + proofs
<code>memory/ept.rs</code>	705	live EPT (L2)
<code>ept_spec + ept_proof</code>	357	notes beside live
<code>frame allocator + spec</code>	362	allocator + ghost set

6.2 Size

Table 3 is a line count on the tree used for this draft, not a re-baseline of the 15,000-line cap. Ghost proofs of record are larger than the live EPT engine. That is expected: the theorem lives in the crate we can host-verify. The cap still applies to the Proven Core modules; most of those rows are not at L3.

6.3 Runtime: lab host versus R640

Table 4 is the split. Cells are contemporaneous serial or host-test evidence, not a wish list. Nested VT-x is a third column because operators (and reviewers) otherwise collapse it into iron.

On the R640: iDRAC virtual media boots the EFI; COM2 archives VMXON, EPT identity over a precise window, unmodified Linux to a shell, then multi-VM probes and VMX-OFF. Secondary Enable-XSAVES was required on Xeon Scalable; without it the guest init hit a stack guard. Firmware SNP/Tcp4 after ExitBootServices is a platform limit of the virtual-floppy path; durable HTTP is a host-owned Broadcom LOM. The SPA can VMLAUNCH a private-EPT SHELL guest and re-enter; that guest is a CPUID stub, not a distro. Guest firmware issued `ATAPI_READ(10)` and booted El Torito CD EFI. Persist-detect stamps on a USB stick are LBAs, not a guest root filesystem.

On nested VT-x we authored guest firmware after third-party `WaitForEvent` forcing faulted. That firmware completed an alpine-extended `setup-disk -m sys` (installer printed “Please reboot”) and reboot-to-disk. Nested virtualization is not the R640. The USB stick used on iron is too small for that distro image. We do not print an install-complete marker from host CI or from nested runs.

6.4 What we refuse to count as a theorem

72-hour soak metrics on a laptop, PDF audit-report templates, and a bring-up spec review are operational gates. They closed on the lab host. They are L0/L1 process, not L3. A future freeze of this argument as a preprint still needs a third-party re-run of the verifier and a written external spec review. Iron distro install is a product gate. It is not required for the ghost theorem; claiming it would be a correctness reject.

Table 4. Split evaluation. Nested VT-x $\neq$ PowerEdge. Host Verus $\neq$ EFI.

Surface	Lab / CI	Nested VT-x	PowerEdge R640
Verus exclusivity (80 / 0)	yes (L3)	—	not claimed
Kani on unsafe	yes (CI pin)	—	not claimed
VMXON + VMEXIT	yes	yes	yes
EPT identity + Linux 6.12 shell	yes	—	yes
Multi-VM probes	L3 <i>N</i> -guest	—	probes on iron
Durable host-NIC HTTP after EBS	lab NIC	—	onboard LOM
SPA launches a SHELL guest	host smoke	—	stub, not a distro
Guest firmware CD EFI (El Torito)	—	CD+virtio	yes
Distro sysinstall + reboot-to-disk	—	yes	open
Persist-detect LBA stamps on USB	—	—	stamps, not a rootfs

7 Experience: Five Lessons that Generalized

EuroSys asks experience papers for lessons that are general and quantitative [20]. The following are the ones we would tell a group starting a verified Type-1 hypervisor in Rust. They are not the theorem. They are why the theorem is the size it is.

7.1 Pin the prover or the proof is a diary

Verus and its SMT backend move. We froze the tagged Verus release 0.2026.07.12 (asset 0b42f4c) with a SHA-256 of the Linux zip in-tree, and forbade CI from fetching latest. The 80/0 line is replayable only against that pin. Groups that verify “on nightly” will not be able to say, a quarter later, whether a new admit is a regression or a toolchain change. Budget pin maintenance as a first-class task, not as hope.

7.2 The UEFI target cannot run SMT

Isolon’s EFI target is UEFI PE/COFF, not a Linux sysroot. Verus wants a host sysroot. We stopped trying to link proofs into the PE/COFF image. The 2,366-line crate lives on the CI host; the 705-line engine lives in the EFI. That split is why L3 and L2 are different columns in Table 4. A single-crate “verified hypervisor” that does not build on the target you ship will still be described as verified by someone; make the packaging match the claim first.

7.3 Do not poke third-party firmware

When guest firmware blocked in WaitForEvent, a tempting fix is to poke its wait object, skip a check, or inject a fault. We did that on a development path. The firmware then page-faulted on a NULL event pointer and entered a dead loop. We do not own that firmware’s invariants; forcing one of them violates the next. The product path is firmware we author, driven only through architected inputs (interrupts, device registers, DMA the spec says is ours). Device emulation remains outside the Proven Core, so a firmware bug is still not an EPT exclusivity theorem—but it is no longer a self-inflicted one. This lesson is independent of Isolon: if you do not own the blob, do not rewrite its control state.

7.4 Firmware TCP is not a management plane

iDRAC virtual floppy advertised PXE/HTTP but not a Tcp4 service binding. Firmware SNP that had accepted a listen before ExitBootServices did not survive it. Durable HTTP on the R640 is a host-owned Broadcom LOM after the hypervisor owns the NIC, not a firmware socket. Quantitative split: PRE-EBS HTTP closed on SNP as a bring-up window; post-EBS HTTP closed only after a native driver. Papers that claim “the hypervisor serves a Web UI” should say which NIC stack and which side of ExitBootServices. In-process REST on a lab executable is a third, still weaker, claim.

7.5 Nested VT-x is not the production server

Nested alpine-extended sysinstall completed: installer printed “Please reboot,” then reboot-to-disk succeeded. That is a real product-path result. It is also the result most likely to be copy-pasted onto the R640 column. We refuse that paste. The USB stick used on iron is too small for that distro image; El Torito CD EFI is not setup-disk. A lab hypervisor that installs Linux under nested VT-x has not demonstrated ISO install on the machine in the rack. Table 4 exists so we cannot “fix” that with prose.

7.6 Negative results we archived

Table 5 lists failures we kept. They are not theorems. They stopped us from claiming theorems we did not have.

The XSAVES miss is the most transferable: a VMCS secondary-control bit that is optional on a laptop CPU and required on the server. Isolation proofs do not mention it. The guest still does not boot. Treat CPU-feature bring-up as L1 evidence, not as a corollary of exclusive ownership.

8 Related Work

Proved kernels and enclaves. seL4 [10, 11, 17] is the landmark machine-checked microkernel: functional correctness, then integrity and information flow. Isolon borrows the discipline—small trusted core, explicit invariants, proofs against a written theorem—and refuses seL4’s scope. We do not prove the scheduler or the device model. CertiKOS [3, 5]

Table 5. Archived negative results. None of these are L3.

Failure	What it blocked
Guest /init stack guard without Enable-XSAVES	Linux shell on Xeon Scalable
Firmware Tcp4 absent on virtual floppy	“HTTP forever” via SNP
SNP listen dead after ExitBootServices	post-EBS firmware UI
WaitForEvent state poke → #PF	third-party firmware as product ISO path
USB persist LBA stamps	calling stamps a root filesystem
USB stick capacity < alpine-extended	iron sysinstall on that stick

builds a certified concurrent kernel with contextual refinement. Komodo [4] verifies a software enclave monitor on ARM TrustZone. Ironclad Apps [6] push verification out to application principals. These systems show isolation *can* be proved. They do not ship as BOOTX64.EFI on iDRAC virtual media.

Hypervisors with a verification story. Hyperkernel [18] pushes a hypervisor toward push-button verification with a simplified interface. XMHF [22] is an extensible hypervisor framework with selected proofs. Verisoft XT applied VCC to Hyper-V components [14]. SeKVM / MicroV work [15, 16] layers formal memory protection onto a commodity KVM. Isolon is the opposite packaging: a new UEFI binary with a tiny proved ghost registry, not a verified overlay on Linux. We do not claim a larger TCB proof than SeKVM; we claim a smaller theorem that we have actually discharged in Verus, plus honest labels on everything else.

Small-TCB virtualization. NOVA [20] and related microhypervisor designs argue for a small TCB in the EuroSys tradition. Isolon’s cut is the Proven Core list in Table 1, not a second kernel in the guest. OKL4-style microvisors [7] similarly shrink the host. Production hypervisors—ESXi [2], Xen [1], KVM [9]—isolate with EPT/NPT and years of testing. We do not compete on feature count. We publish the gaps (live EPT at L2, iron installer open) instead of implying the theorem covers the SPA.

Tooling. Verus [12, 13] is the SMT prover we pin. Kani [21] covers unsafe. Push-button verification of file systems [19] is the methodological cousin of Hyperkernel; our exclusivity proofs are lemma-heavy rather than fully push-button across the EFI.

9 Limitations

- Live executable EPT is L2. L3 is the host-only ghost model. Ghost↔exec refine is scoped, not a full EFI bisimulation.
- Most Proven Core modules in Table 1 are not at L3.

- Large-page and NUMA L3 are lab/CI. Iron stopped at multi-VM probes for those surfaces.
- PageTransferStep is a ghost primitive, not live migration product.
- DMA confinement, IOMMU, and untrusted-device memory are future work.
- Nested distro sysinstall is complete and is not iron.
- Iron installer, TLS, and guest console remain product residuals.
- Verus upgrades can break proofs; we budget pin maintenance and refuse latest.

Threats to validity. Iron evidence is one PowerEdge R640 and iDRAC virtual media, not a fleet. Nested VT-x is one lab host. The 80/0 line is one pin of one prover; we have not replayed under a second SMT solver. The spec review was a bring-up self-review, not an external auditor. LOC counts are wc -l, not a proven TCB. Experience lessons (XSAVES, SNP, firmware pokes) are one implementation’s failures; we argue they generalize because they follow from Intel VMX, UEFI ExitBootServices, and “do not own that blob,” not from Isolon-specific APIs. We may be wrong about that generalization.

10 Conclusion

Isolon now has what a living verification argument is supposed to wait for before *writing* the claim, not before *freezing* it.

On the proof side: exclusive ownership is stated tightly, implemented as a ghost model, and discharged through map, unmap, violation, and page transfer under a frozen Verus pin—80 verified, 0 errors—with a spec review that asked “are we proving the right thing?” out loud and did not find a proxy theorem. That is a verified *core model*. It is not a verified .efi.

On the iron side: a PowerEdge R640 printed VMX, EPT, Linux shell, host-NIC HTTP, a SHELL guest from the SPA, and El Torito CD EFI. Nested authored firmware completed a distro sysinstall. Nested is not iron. The remaining product sentence is the same installer on metal, on a stick that fits the image, without claiming a marker we have not printed.

Memory isolation isn’t tested. The ghost exclusivity theorems are proved. The product loop is not done. Those three sentences are the paper.

References

- [1] Paul Barham, Boris Dragovic, Keir Fraser, Steven Hand, Tim Harris, Alex Ho, Rolf Neugebauer, Ian Pratt, and Andrew Warfield. 2003. Xen and the Art of Virtualization. In *Proceedings of the 19th ACM Symposium on Operating Systems Principles (SOSP)*. 164–177. <https://doi.org/10.1145/945445.945462>
- [2] Edouard Bugnion, Scott Devine, Mendel Rosenblum, Jeremy Sugerman, and Edward M. Bugnion. 2012. Bringing Virtualization to the x86 Architecture with the Original VMware Workstation. *ACM*

Table 6. Where Isolon sits. “Proved” means a machine-checked theorem of the named property, not “tests passed.”

System	What is proved	Guest / boot	Isolation property	This paper
seL4	kernel func. + integrity/IFC	user tasks, not UEFI HV	info-flow / integrity	discipline, not scope
CertiKOS	concurrent kernel refine	certified OS	contextual refine	not our TCB cut
Komodo	enclave monitor	TrustZone	enclave isolation	different hardware
Hyperkernel	push-button kernel	simplified HV iface	interface properties	we keep full VMX/EPT
SeKVM	KVM memory protection	Linux+KVM	hyp. memory protection	overlay vs clean-slate
NOVA	(design) small TCB	microhypervisor	TCB size argument	our cut is Proven Core
Xen/KVM/ESXi	—	production guests	tested EPT/NPT	we claim a smaller theorem
Isolon	EPT exclusive ownership (ghost)	UEFI .efi, Linux on R640	exclusive frame ownership	L3 ghost, L2 live EPT

- Transactions on Computer Systems* 30, 4 (2012), 12:1–12:51. <https://doi.org/10.1145/2384592.2384593>
- [3] David Costanzo, Zhong Shao, and Ronghui Gu. 2016. End-to-End Verification of Information-Flow Security for C and Assembly Programs. In *Proceedings of the 37th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*. 648–664. <https://doi.org/10.1145/2908080.2908100>
- [4] Andrew Ferraiuolo, Andrew Baumann, Chris Hawblitzel, and Bryan Parno. 2017. Komodo: Using Verification to Disentangle Secure-Enclave Hardware from Software. In *Proceedings of the 26th ACM Symposium on Operating Systems Principles (SOSP)*. 287–305. <https://doi.org/10.1145/3132747.3132782>
- [5] Ronghui Gu, Zhong Shao, Hao Chen, Xiongnan (Newman) Wu, Jieung Kim, Vilhelm Sjöberg, and David Costanzo. 2016. CertiKOS: An Extensible Architecture for Building Certified Concurrent OS Kernels. In *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 653–669.
- [6] Chris Hawblitzel, Jon Howell, Jacob R. Lorch, Arjun Narayan, Bryan Parno, Danfeng Zhang, and Brian Zill. 2013. Ironclad Apps: End-to-End Security via Automated Full-System Verification. In *Proceedings of the 11th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 165–181.
- [7] Gernot Heiser and Ben Leslie. 2010. The OKL4 Microvisor: Convergence Point of Microkernels and Hypervisors. In *Proceedings of the First ACM Asia-Pacific Workshop on Systems (APSys)*. 19–24. <https://doi.org/10.1145/1851276.1851282>
- [8] Intel Corporation. 2024. Intel 64 and IA-32 Architectures Software Developer’s Manual, Volume 3: System Programming Guide. (2024). Order number 325384. Extended page tables (EPT) and VMX.
- [9] Avi Kivity, Yaniv Kamay, Dor Laor, Uri Lublin, and Anthony Liguori. 2007. KVM: The Linux Virtual Machine Monitor. In *Proceedings of the Linux Symposium*. 225–230.
- [10] Gerwin Klein, June Andronick, Kevin Elphinstone, Toby Murray, Thomas Sewell, Rafal Kolanski, and Gernot Heiser. 2014. Comprehensive Formal Verification of an OS Microkernel. *ACM Transactions on Computer Systems* 32, 1 (2014), 2:1–2:70. <https://doi.org/10.1145/2560537>
- [11] Gerwin Klein, Kevin Elphinstone, Gernot Heiser, June Andronick, David Cock, Philip Derrin, Dhammika Elkaduwe, Kai Engelhardt, Rafal Kolanski, Michael Norrish, Thomas Sewell, Harvey Tuch, and Simon Winwood. 2009. seL4: Formal Verification of an OS Kernel. In *Proceedings of the ACM SIGOPS 22nd Symposium on Operating Systems Principles (SOSP)*. 207–220. <https://doi.org/10.1145/1629575.1629596>
- [12] Andrea Lattuada, Travis Hance, Jay Bosamiya, Matthias Brun, Chanhee Cho, Hayley LeBlanc, Pranav Srinivasan, Reto Achermann, Tej Chajed, Chris Hawblitzel, Jon Howell, Jacob R. Lorch, Oded Padon, and Bryan Parno. 2024. Verus: A Practical Foundation for Systems Verification. In *Proceedings of the ACM SIGOPS 30th Symposium on Operating Systems Principles (SOSP)*. 438–454. <https://doi.org/10.1145/3694715.3695952>
- [13] Andrea Lattuada, Travis Hance, Chanhee Cho, Matthias Brun, Isitha Subasinghe, Yi Zhou, Jon Howell, Bryan Parno, and Chris Hawblitzel. 2023. Verus: Verifying Rust Programs using Linear Ghost Types. In *Proceedings of the ACM on Programming Languages*, Vol. 7. 85:1–85:30. <https://doi.org/10.1145/3586037>
- [14] Dirk Leinenbach and Thomas Santen. 2009. Verifying the Microsoft Hyper-V Hypervisor with VCC. In *Proceedings of the 2nd World Congress on Formal Methods (FM)*. 806–809. https://doi.org/10.1007/978-3-642-05089-5_51
- [15] Shih-Wei Li, Xupeng Li, Ronghui Gu, Jason Nieh, and John Zhuang Hui. 2021. Formally Verified Memory Protection for a Commodity Multiprocessor Hypervisor. In *Proceedings of the 30th USENIX Security Symposium*. 3953–3970.
- [16] Shih-Wei Li, Xupeng Li, Ronghui Gu, Jason Nieh, and John Zhuang Hui. 2021. A Secure and Formally Verified Linux KVM Hypervisor. In *Proceedings of the 2021 IEEE Symposium on Security and Privacy (SP)*. 1782–1799. <https://doi.org/10.1109/SP40001.2021.00049>
- [17] Toby Murray, Daniel Matichuk, Matthew Brassil, Peter Gammie, Timothy Bourke, Sean Seefried, Corey Lewis, Xin Gao, and Gerwin Klein. 2013. seL4: From General Purpose to a Proof of Information Flow Enforcement. In *Proceedings of the 2013 IEEE Symposium on Security and Privacy*. 415–429. <https://doi.org/10.1109/SP.2013.35>
- [18] Luke Nelson, Helgi Sigurbjarnarson, Kaiyuan Zhang, Dylan Johnson, James Bornholt, Emina Torlak, and Xi Wang. 2017. Hyperkernel: Push-Button Verification of an OS Kernel. In *Proceedings of the 26th ACM Symposium on Operating Systems Principles (SOSP)*. 252–269. <https://doi.org/10.1145/3132747.3132748>
- [19] Helgi Sigurbjarnarson, James Bornholt, Emina Torlak, and Xi Wang. 2016. Push-Button Verification of File Systems via Crash Refinement. In *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 1–16.
- [20] Udo Steinberg and Bernhard Kauer. 2010. NOVA: A Microhypervisor-Based Secure Virtualization Architecture. In *Proceedings of the 5th European Conference on Computer Systems (EuroSys)*. 209–222. <https://doi.org/10.1145/1755913.1755935>
- [21] Alexa VanHattum, Daniel Schwartz-Narbonne, Nathan Chong, and Adrian Sampson. 2022. Verifying Dynamic Trait Objects in Rust. In *Proceedings of the 44th International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP)*. 321–330. <https://doi.org/10.1145/3510457.3513039>
- [22] Amit Vasudevan, Sagar Chaki, Limin Jia, Jonathan McCune, James Newsome, and Anupam Datta. 2013. Design, Implementation and Verification of an Extensible and Modular Hypervisor Framework. In *Proceedings of the 2013 IEEE Symposium on Security and Privacy*. 430–444. <https://doi.org/10.1109/SP.2013.36>